9-Oct-26.

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Subject : PF lab.

Class CS Sec E

"Assignment lab 2"

Task 1:- Type and save the following programs in visual studio. Run these programs and observe their output.

(a):-

Input:-

```
#include <iostream>
Using namespace std;
int main()
{
```

```
Char   letter;
letter  = 'A';
Cout << letter <<'\n';
letter = 'B';
cout << letter <<'\n';
Cin >> letter;
return 0;
}
```

Output:-

A
B
57

(b):- Input:-

```
#include <iostream>
Using namespace std;
int main()
{
   Char  letter;
   letter = 65;
   Cout << letter <<'\n';
   letter = 66;
   Cout << letter <<'\n';
   Cin >> letter;
   return 0;
}
```

Output:-

A
B
86.

10:14 PM

new*

```cpp
#include<iostream>
using namespace std;

int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin>>letter;

    return 0;
}
```

Tab | { | } | : | ; | "

10:14 PM

new*

```cpp
#include<iostream>
using namespace std;

int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin>>letter;

    return 0;
}
```

Tab | { | } | : | ; | "

A

B

A

B

(C):- Input:-

```
#include <iostream>
Using namespace std;
int main()
{
    int letter;
 letter = 'A';
cout << letter << '\n';
 letter = 'B';
cout << letter <<'\n';
Cin >> letter;
return 0;
}
```

Output:-

```
65
66.
```

Task 2:- Take two integers as input from the user apply authematic operators on them (+,-,*,/) as print them on

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new*

```cpp
#include<iostream>
using namespace std;

int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin>>letter;

    return 0;
}
```

Tab | { | } | : | ; | "




```
65
66
...

[Program finished]
```

Output:-

```
#include <iostream>
Using (main) namespace std;
int main()
{
   int a,b;
  Cin >> a >> b;
  Cout << a+b << endl;
  Cout << a-b << endl;
  Cout << a *b<<endl;
  Cout <<(float) a /b<<endl;
   return 0;
}
```

Output:-

```
127
-127
 0
 0
```

Task: 03

Wite the output of the following code.

```
# include "stda fx.h"
#include < iostream>
 using namespace std;
 int main ( ) {
int val 1;
int val 2;
int val 3;
Cout <<"please enter your 3 numbers:";
 Cin >> val 1 >>val 2 >> Val 3;
Cout <<val 2 <<"\n"<<val 2<< "\n"<<val3<<"\n";
Cout <<val 3 << "\n"<<val 2<<"\n"<<val<<"\n";
   System ("pause");
   return 0;
}
```

Output:

```
please enter your 3 numbers: 1 2 3
1
2
3
3
2
1
```

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new*

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;

    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    return 0;
}
```

Tab | { | } | : | ; | "

```
please enter your 3 numbers: 1 2 3
1
2
3
3
2
1
```

Task4 :-

Write a C++ program to calculate the distance between the two points. Write Note: x1, y1, x2, y2 are all double values.

Formula $d = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Input :-

```
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    double x1, y1, x2, y2;
    cout << "Enter x1 y1:";
    cin >> x1 >> y1;
    cout << "Enter x2 y2:";
    cin >> x2 >> y2;
    double dx = x2 - x1;
```

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new*

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    double x1, y1, x2, y2;

    cout << "Enter x1 y1: ";
    cin >> x1 >> y1;

    cout << "Enter x2 y2: ";
    cin >> x2 >> y2;

    double dx = x2 - x1;
    double dy = y2 - y1;
    double d = sqrt(dx*dx + dy*dy);

    cout << "Distance = " << d;

    return 0;
}
```

Tab | { | } | : | ; | "

```
Enter x1 y1 x2 y2: 0 0 3 4
Distance = 5
```

Output:-

Please enter your 3 numbers : 678.

```
424
789
678
424
789
789
```

```
424
578.
```

Task 5:-

1:- Calculate the area of a circle (area = PI * r2)

Input:-

(a):-

```
#include <iostream>
using namespace std;
int main()
{
    float r, CircleArea;
    const float PI = 3.14159;
    cout << "Enter radius of
    circle: ";
```

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new*

```cpp
#include <iostream>
using namespace std;

int main() {
    float r, area;
    const float PI = 3.14159;
    cout << "Enter radius: ";
    cin >> r;
    area = PI * r * r;
    cout << "Area of Circle = " << area;
    return 0;
}
```

Tab | { | } | : | ; | "

```
Enter x1 y1: 68
7828
Enter x2 y2: 7738
6617
Distance = 7765.01
[Program finished]
```

```
cin >> r;
area = PI * r * r;
cout << "Area of circle = " << area;
return 0;
}
```

```
Enter radius: 96
Area of circle =
28952.9.
```

(b):-

2]:- Calculate the area of a Rectangle (area = length * width)

Input:-

```
#include <iostream>
Using namespace std;
int main()
{ float length, width, area;
cout << Enter length: ";
cin >> length;
cout << Enter width:";
cin >> width;
```

8:47

October 9, 2026
8:30 | WhatsApp Business

```cpp
#include <iostream>
using namespace std;

int main() {
    float length, width, area;
    cout << "Enter length: ";
    cin >> length;
    cout << "Enter width: ";
    cin >> width;
    area = length * width;
    cout << "Area of Rectangle = " << area;
    return 0;
}
```




```
Enter length: 46
Enter width: 86
Area of Rectangle = 3956
[Program finished]
```

```
area = length * width;
Cout<< area of Rectangle=" <<area;                    dy);
 return 0;

}
```

Output:-

```
Enter length:
16
Enter width:
86
Area of Recta
= 3956.
```

(C):- Calculate the area of a triangle (area = base * hight * .5)

Input:-

```
#include <iostream>
Using namespace std;
int main()
{
   float base, height, area;
   Cout << "Enter base" ;
   Cin >> base;
   Cout << "Enter height:" ;
   Cin >> height;
   Cout <<
area = base * height * 0.5;
```

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new*

```cpp
#include <iostream>
using namespace std;

int main() {
    float base, height, area;
    cout << "Enter base: ";
    cin >> base;
    cout << "Enter height: ";
    cin >> height;
    area = base * height * 0.5;
    cout << "Area of Triangle = " << area;
    return 0;
}
```

Tab | { | } | : | ; | "

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TAB

```
Enter base: 67
Enter height: 53
Area of Triangle = 1775.5
[Program finished]
```

cout << area of triangle = " <<
area;
return 0;
}

```
Enter base: 67
Enter height: 53
Area of Triangle
= 1775.5.
```

y);

Task 6:-

```
#include <iostream>
Using namespace std;
int main()
{
    double i1, 12.95, i2 = 24.95, i3 = 6.95,
i4 = 14.95, i5, 3.95;
double Sub = i1 + i2 + i3 + i4 + i5;
    cout << i1 << endl << i2 << endl <<
i3 << endl << i4 << endl << i5 << endl;
    cout << "Subtotal: " << sub << \n Tax: "
<< Sub * 0.06 << "\n Total: " << sub * 1.06;
}
```

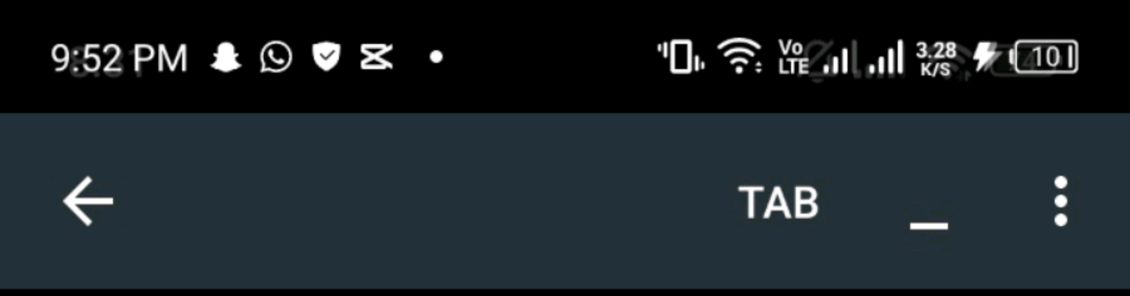
```
12.95   | Subtotal:
24.95   | 63.75
5.95    | Tax: 3.825
14.95   | Total:
3.95    | 67.575.
```

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new*

```cpp
#include <iostream>
using namespace std;
int main() {
    double i1=12.95, i2=24.95, i3=6.95, i4=14.95, i5=3.95;
    double sub = i1+i2+i3+i4+i5;
    cout << i1 << endl << i2 << endl << i3 << endl << i4 << endl << i5 << endl;
    cout << "Subtotal: " << sub << "\nTax: " << sub*0.06 << "\nTotal: " << sub*1.06;
}
```

Tab | { | } | : | ; | "




```
12.95
24.95
6.95
14.95
3.95
Subtotal: 63.75
Tax: 3.825
Total: 67.575
[Program finished]
```